

Inherited Causes of Nephrotic Syndrome

Ronen Schneider, Friedhelm Hildebrandt

INTRODUCTION

The discovery of the genes encoding the slit membrane proteins nephrin (*NPHS1*),¹ podocin (*NPHS2*),² and phospholipase C epsilon 1 (*PLCE1*)³ as causing steroid-resistant nephrotic syndrome (NS), if mutated, has stimulated extensive research into monogenic causes of NS. In the last 20 years, acceleration of next-generation sequencing has permitted identification of more than 63 monogenic causes of NS, and this number is rapidly increasing (Fig. 20.1 and references in Table 20.1). Most of these genes are highly expressed in the glomerular podocytes, implicating podocytes as the primary site of injury in NS. The encoded proteins of these genes cluster within functional pathways critical for podocyte biology and cause disruption of the filtration barrier if mutated.

Epidemiology

Steroid-resistant nephrotic syndrome (SRNS) is the second leading cause of pediatric chronic kidney disease (CKD), responsible for 15% of all CKD cases before the age of 25 years. A monogenic cause of SRNS can be found in 10% to 30% of cases. Specific SRNS genes or distinct mutations (alleles) in the same SRNS gene may cause characteristic phenotypes, which may pertain to age of onset of disease.^{4,5} Adult-onset NS is of heterogeneous origin and may include causes such as membranous glomerulonephritis, minimal change nephropathy, diabetic nephropathy, and amyloid nephropathy. The majority of these diseases do not seem to have a genetic basis, at least not a monogenic one, and genetic workup for polymorphisms in these diseases (e.g., *PLA2R* polymorphisms) does not have clinical relevance so far. However, genetic testing should be considered in selected cases of adult-onset focal segmental glomerulosclerosis (FSGS) (see “Diagnosis” section).

A study using next-generation sequencing tested a large international cohort of patients with SRNS manifesting before age 25 years (1783 families).⁴ The diagnostic panel included 21 genes with a recessive mode of inheritance and six genes with a dominant mode of inheritance. A genetic diagnosis was established in 526 of 1783 families (29.5%) in whom the condition manifested before age 25 years.⁴ Stringent criteria for calling mutations “disease causing” were applied (Box 20.1). The highest rate of mutation detection of 69% was in the youngest group of patients (0–3 months), a percentage that decreased with older age (Fig. 20.2).

A recent study used whole-exome sequencing (WES) analysis and established a genetic diagnosis in 25% of individuals younger than 25 years who had SRNS.⁶ This study showed that four genes were major SRNS genes: *NPHS1* (4.3% of 300 individuals with SRNS), *PLCE1* (3.7%), *NPHS2* (2.7%), and *SMARCA1* (2.7%) (Fig. 20.3). The most prevalent SRNS dominant genes are *INF2*, *TRPC6*, and *ACTN4* (see Fig. 20.3 and Table 20.1).⁶

Etiology

The proteins encoded by the genes that cause monogenic SRNS map back onto distinct structural protein complexes and signaling pathways that reveal what is essential for glomerular function (see Fig. 20.1). In particular, it has been determined that almost all mutations involve genes/proteins important for the slit diaphragm function or essential pathways in podocytes. Interdigitations of podocyte foot processes form the glomerular slit membrane between them, which is critical for the filtering process and retention of protein in the bloodstream (see also Chapter 1). The integrity of the glomerular slit membrane is lost in NS. Thus, identification of monogenic causes of NS revealed dozens of proteins, each of which is an indispensable component of glomerular function, because loss of their function in a monogenic form of NS inescapably leads to proteinuria and FSGS. Knowledge of these mutations helps to understand the physiology of the glomerular filtration barrier and the pathogenic mechanisms which cause NS. These functional complexes include proteins involved in glomerular slit membrane-associated components, actin-binding proteins, actin-regulating small GTPases of the Rho/Rac/Cdc42 family, coenzyme Q₁₀ (CoQ₁₀) biosynthesis, sphingosine-1-phosphate (S1P) metabolism, nuclear transcription factors, nucleoporins, lysosomal proteins, endosomal trafficking regulating proteins, tRNA modification (KEOPS complex), and laminin/integrin receptors (focal adhesions) (see Fig. 20.1).

The power of identification of such single-gene causes of NS lies in the fact that recessive mutations (and to a lesser degree, dominant mutations) almost always convey full penetrance and thereby represent the cause of NS (i.e., the etiology) rather than representing only an increased risk for acquiring disease (see Box 20.1). Thus, mutations in the 63 monogenic genes can be considered causative of NS (see Table 20.1).

CLINICAL PRESENTATION

Mutations in recessive genes tend to cause SRNS in early childhood, whereas there is a tendency for mutations in dominant SRNS genes (e.g., *TRPC6*, *INF2*) to cause adult-onset SRNS,^{7,8} with the exception of *WT1*.⁴ In a study, mutations in *WT1* have been shown to exhibit a biphasic distribution for onset of NS with a first peak at 4 to 12 months and a second peak for age of onset beyond 18 years of age (see Fig. 20.2).⁴ This difference in age of presentation between recessive and dominant diseases is a well-reported genetic phenomenon that is not completely understood. Perhaps late onset could be explained by molecular mechanisms in dominant diseases that can result in an age-related penetrance. These mechanisms include haploinsufficiency, gain-of-function, and dominant-negative effects that often do not

Proteins That, if Mutated, Can Cause Monogenic Steroid-Resistant Nephrotic Syndrome

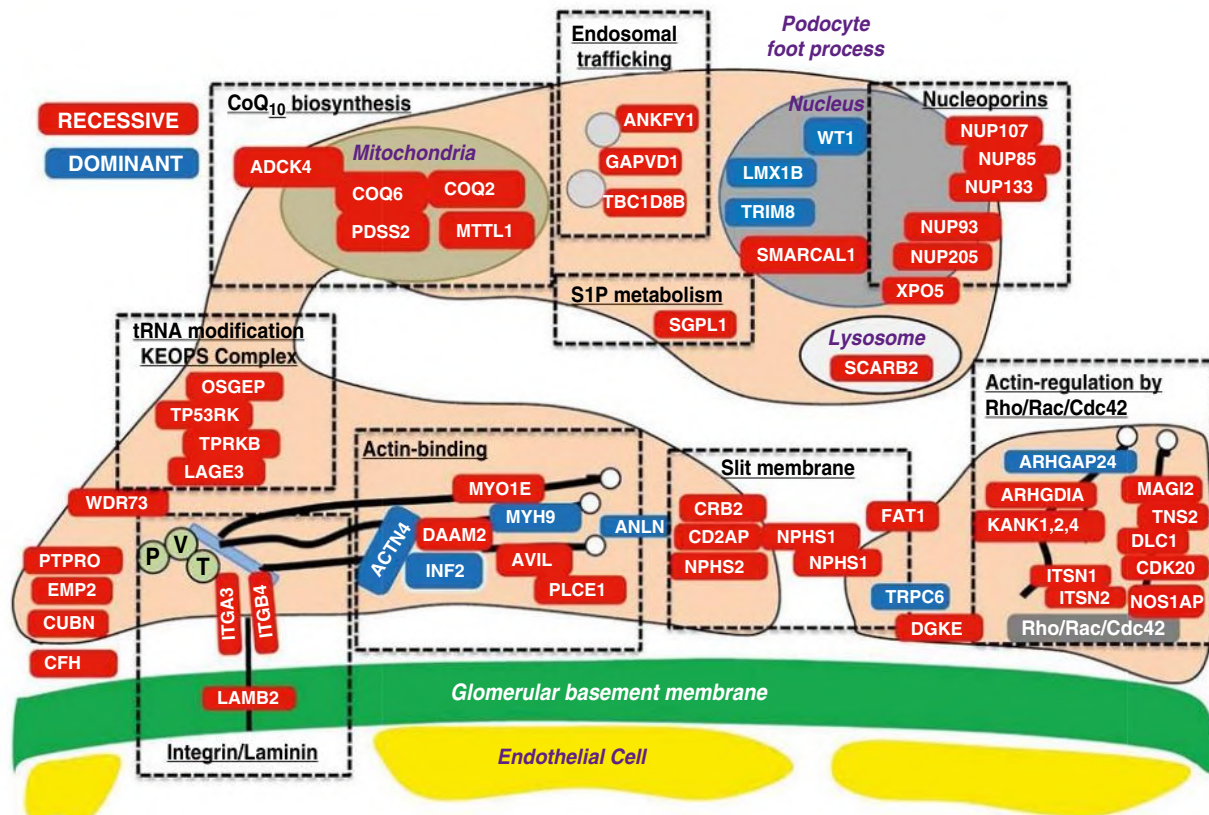


Fig. 20.1 Proteins Whose Genes, if Mutated, Cause Monogenic Steroid-Resistant Nephrotic Syndrome (SRNS) Define Essential Podocyte Functions. Identification of 63 single-gene (monogenic) causes of SRNS has revealed the glomerular epithelial cell, the podocyte, as central to the pathogenesis of SRNS, because almost all of the related genes are relevantly expressed in podocytes. Identification of genes that, if mutated, cause SRNS revealed distinct proteins and functional pathways as essential for glomerular function, because a mutation in any single one of them is sufficient to cause SRNS. This figure depicts a simplified cross section through a podocyte (*tan*) and a neighboring podocyte foot process, which is attached to the glomerular basement membrane (*green*) by laminin-integrin receptors. Fifty-three proteins whose genes, if mutated, cause recessive monogenic forms of SRNS are shown in *red*, and 10 proteins whose genes, if mutated, cause dominant forms of SRNS are shown in *blue* (see [Tables 20.1](#) and [20.2](#)). The 63 SRNS-related proteins were found to be part of protein-protein interaction complexes that participate in defined structural components or signaling pathways of podocyte function (shown within *black dashed boxes*). These proteins include proteins involved in coenzyme Q10 biosynthesis, S1P metabolism, nuclear transcription factors, nucleoporins, lysosomal proteins, actin-regulating small GTPases of the Rho/Rac/Cdc42 family, glomerular slit membrane-associated components, actin-binding proteins, tRNA modification (KEOPS complex), endosomal trafficking regulators, and laminin/integrin receptors (focal adhesions). Proteins that are encoded by recessive NS genes are marked in *red*. *ADCK4*, aarF domain containing kinase 4; *ANKFY1*, Rabankyrin-5; *ARHGDIA*, Rho GDP dissociation inhibitor (GDI) α ; *AVIL*, advillin; *CD2AP*, CD2-associated protein; *CDK20*, cyclin-dependent kinase 20; *COQ2*, coenzyme Q2 4-hydroxybenzoate polyprenyltransferase; *COQ6*, coenzyme Q10 monooxygenase 6; *CRB2*, crumbs family member 2; *CUBN*, cubilin (intrinsic factor-cobalamin receptor); *DAAM2*, disheveled-associated activator of morphogenesis 2; *DGKE*, diacylglycerol kinase; *DLC1*, deleted in liver cancer 1; *EMP2*, epithelial membrane protein 2; *FAT1*, FAT tumor suppressor homolog 1; *GAPVD1*, GTPase-activating protein and VPS9 domain-containing protein 1; *GON7*, EKC/KEOPS complex subunit GON7; *ITGA3*, integrin, $\alpha 3$; *ITGB4*, integrin, $\beta 4$; *ITSN1*, intersectin 1; *ITSN2*, intersectin 2; *KANK*, KN motif and ankyrin repeat domains 1/2/4; *LAGE3*, L antigen family member 3; *LAMB2*, laminin, $\beta 2$; *MAGI2*, membrane-associated guanylate kinase, WW and PDZ domain containing 2; *MTTL1*, mitochondrial tRNA leucine 1; *MYO1E*, Homo sapiens myosin 1e; *NOS1AP*, Carboxyl-terminal PDZ ligand of neuronal nitric oxide synthase protein; *NPHS1*, nephrin; *NPHS2*, podocin; *NUP85*, nucleoporin 85 kDa; *NUP93*, nucleoporin 93 kDa; *NUP107*, nucleoporin 107 kDa; *NUP133*, nucleoporin 133 kDa; *NUP205*, nucleoporin 205 kDa; *OSGEP*, O-sialoglycoprotein endopeptidase; *PDSS2*, prenyl (decaprenyl) diphosphate synthase, subunit 2; *PLCE1*, phospholipase C, ϵ 1; *PTPRO*, protein tyrosine phosphatase, receptor type O; *SCARB2*, scavenger receptor class B member 2; *SGPL1*, sphingosine-1-phosphate lyase 1; *SMARCAL1*, SWI/SNF-related, matrix-associated, actin-dependent regulator of chromatin, subfamily a-like 1; *TBC1D8B*, TBC1 domain family member 8B; *TNS2*, tensin 2; *TP53RK*, TP53-regulating kinase; *TPRKB*, TP53RK-binding protein; *WDR73*, WD repeat domain 73; and *XPO5*, Exportin; *YRDC*, YrdC domain-containing protein, mitochondrial 5. Proteins that are encoded by dominant NS genes are marked in *blue*: *ACTN4*, actinin, $\alpha 4$; *ANLN*, anillin; *ARHGAP24*, Rho GTPase-activating protein 24; *INF2*, inverted formin, FH2 and WH2 domain containing; *LMX1B*, LIM homeobox transcription factor 1- β ; *MYH9*, myosin, heavy chain 9; *TRIM8*, E3 ubiquitin-protein ligase TRIM8; *TRPC6*, transient receptor potential cation channel, subfamily C, member 6; *WT1*, Wilms tumor 1. (Modified from Lovric S, Ashraf S, Tan W, Hildebrandt F. Genetic testing in steroid-resistant nephrotic syndrome: when and how? *Nephrol Dial Transplant*. 2016;31(11):1802–1813.)

TABLE 20.1 Genes That, if Mutated, Cause Monogenic Steroid-Resistant, Steroid-Sensitive, or Dependent Nephrotic Syndrome^a

Gene	Mode of Inheritance	Clinical Aspects	Reference
Genetic Causes of Nephrotic Syndrome That First Manifest in Adolescents and Adults			
<i>INF2</i> ^b	AD	Potential neurologic manifestations (Charcot-Marie-Tooth syndrome, see Table 20.2)	8
<i>WT1</i> ^b	AD	Biphasic age of onset Risk for WT/gonadoblastoma (mutation dependent, see Table 20.2)	39
<i>ACTN4</i> ^b	AD	Screen for mutation in family members if kidney donation is considered	40
<i>TRPC6</i>	AD	Familial FSGS, screen for mutation in family members if kidney donation is considered NS onset can present in up to fifth decade of life	7
<i>MYH9</i>	AD	Screen for mutation in family members if kidney donation is considered	41
<i>PAX2</i>	AD	Screen for mutation in family members if kidney donation is considered Variable age of presentation (2–20 years)	42
<i>ANLN</i>	AD	Screen for mutation in family members if kidney donation is considered	43
Genetic Causes of Nephrotic Syndrome That First Manifest in Infancy to Early Childhood			
<i>NPHS</i> ^c	AR	Congenital nephrotic syndrome Common in Finland Infants born prematurely with massive proteinuria and often have low-set ears and a small nose Mutation-dependent prognosis; ESKD in most cases by age 3 years	1
<i>NPHS2</i> ^c	AR	Early childhood or adolescent-adult onset, mutation dependent	2
<i>PLCE1</i> ^c	AR	Incomplete penetrance was observed (very rare for AR disease)	3
<i>LAMB2</i> ^c	AR	Potential eye abnormalities (Pierson syndrome; see Table 20.2)	44
<i>SMARCAL1</i> ^c	AR	Potential skeletal abnormalities (Schimke immuno-osseous dysplasia; see Table 20.2)	45
<i>LMX1B</i>	AD	See Table 20.2	46
<i>ARHGDI</i>	AR	May be responsive to eplerenone treatment	33
<i>CDK20</i>	AR	May respond to steroid treatment	15
<i>COQ2, COQ6, COQ8/ADCK4, PDSS2</i>	AR	May respond to CoQ10 supplementation	10, 19, 47
<i>CUBN</i>	AR	Can manifest as proteinuria with no progression to ESKD; may respond to treatment with vitamin B ₁₂	31
<i>DGKE</i>	AR	Can present as SRNS in first decade of life or as an acute episode of aHUS during first year of life	48
<i>DLC1</i>	AR	May respond to steroid treatment	15
<i>EMP2</i>	AR	May respond to steroid treatment	20
<i>FAT1</i>	AR	Potential syndromic features (allele dependent): colobomatous microphthalmia, ptosis, syndactyly	49, 50
<i>ITSN1</i>	AR	May respond to steroid treatment	15
<i>ITSN2</i>	AR	May respond to steroid treatment	15
<i>MAGI2</i>	AR	May respond to steroid treatment	51
<i>TNS2</i>	AR	May respond to steroid treatment	15
<i>TRIM8</i>	AD	All mutations described thus far are de novo Extrarenal manifestation: epilepsy	52

^aGenes for which no clear clinical clues to diagnosis were described, or which are too rare as monogenic SRNS to have clinical considerations concluded, are not shown.

^bMost common dominant SRNS causing genes.

^cMost common recessive SRNS causing genes.

AD, Autosomal dominant; aHUS, atypical hemolytic uremic syndrome; AR, autosomal recessive; CoQ, coenzyme Q; ESKD, end-stage kidney disease; FSGS, focal segmental glomerulosclerosis; NS, nephrotic syndrome; SRNS, steroid-resistant nephrotic syndrome; WT, Wilms tumor.

result in a complete loss of function of the encoded proteins, as in recessive diseases.

SRNS manifests histologically as FSGS, a lesion characterized by segmental sclerosis often with diffuse podocyte foot process effacement.⁹ Diffuse mesangial sclerosis (DMS) is a pathogenic finding that

may occur in patients with early-onset NS. In DMS, light microscopy shows small, sclerosed glomeruli with a reduced number of capillary loops together with podocyte hypertrophy and mesangial matrix expansion. SRNS may rarely be associated with extrarenal manifestations such as neurologic defects (Table 20.2).¹⁰

BOX 20.1 Assignment of Mutations as Being Disease Causing

Basic Assumptions

- Defined clinical phenotype
- Known genes with similar phenotype have been excluded
- “Mutation” implies that an allele changes the phenotype
- Full (AR) or incomplete (AD) penetrance (age related)

Include Allele as Disease Causing if:

- Truncating mutation (stop, abrogation of start or stop, obligatory splice, frameshift) in an expressed gene (well-annotated mRNA, conservation, protein expression), *or*
- Missense mutation if:
 - Continuously conserved at least up to *Danio rerio* (zebrafish), *and*
 - Loss of function in human allele is supported by functional data, *and*
 - Allele segregates with the affected status in the family
 - Known genes with similar phenotype have been excluded

Exclude Allele as Disease Causing if:

- Heterozygous allele frequency >1% (for AR) or >0.1% (for AD) or reported homozygous in gnomAD (Genome Aggregation Database) in AR disease, or frequent heterozygously (>20) in AD disease
- AR: nonsegregation (e.g., “compound heterozygous” in cis; affected family member is without the variant; unaffected parent is with homozygous variant)
- AD: nonsegregation (e.g., affected family member is without the variant; but if an affected family member is with the allele, consider incomplete penetrance and variable expressivity)

AD, Autosomal dominant; AR, autosomal recessive.

Disease presentation of NS may vary in a gene-specific or even allele-specific manner (i.e., specific mutations within the same gene may determine a phenotypic range) that can affect age of onset, severity of disease, response to steroids, and associated syndromic features. For instance, mutations in *NPHS1* encoding nephrin will most often result in congenital nephrotic syndrome (CNS) characterized by massive proteinuria not responding to steroids and will likely require kidney replacement therapy and bilateral nephrectomy at approximately 1 year of age. Until permissive body weight (>10 kg) is achieved to enable kidney transplantation, these infants are at a high risk of life-threatening complications. However, specific mutations, such as the introduction of a stop codon (non-sense mutation) at position p.1160 of nephrin, tend to cause a milder form of CNS that may respond to steroid treatment.¹¹ In *NPHS2* (encoding podocin) mutations, which are the most common causative mutations found in monogenic SRNS, the mutation R138Q causes onset in early childhood,¹² whereas the mutation R229Q in compound heterozygosity with specific second mutations causes later-onset SRNS.¹³ With *INF2*, a heterozygous mutation will typically result in SRNS and FSGS with adolescent- or adult-onset disease, with or without neurologic manifestations (Charcot-Marie-Tooth syndrome; see Table 20.2).¹⁴ Mutations in recessive genes that cause SRNS/FSGS with syndromic features (see Table 20.2) tend to be of early onset, severe, and carry a poor prognosis. On the other hand, mutations in one of six recently discovered genes may result in partial SSNS¹⁵ (see Table 20.1).

DIAGNOSIS

Next-generation sequencing of a specified panel of genes, such as an SRNS-disease genes panel, or WES of mutations across the entire

Relative Distribution of Nephrotic Syndrome—Causative Genes by Age of Onset

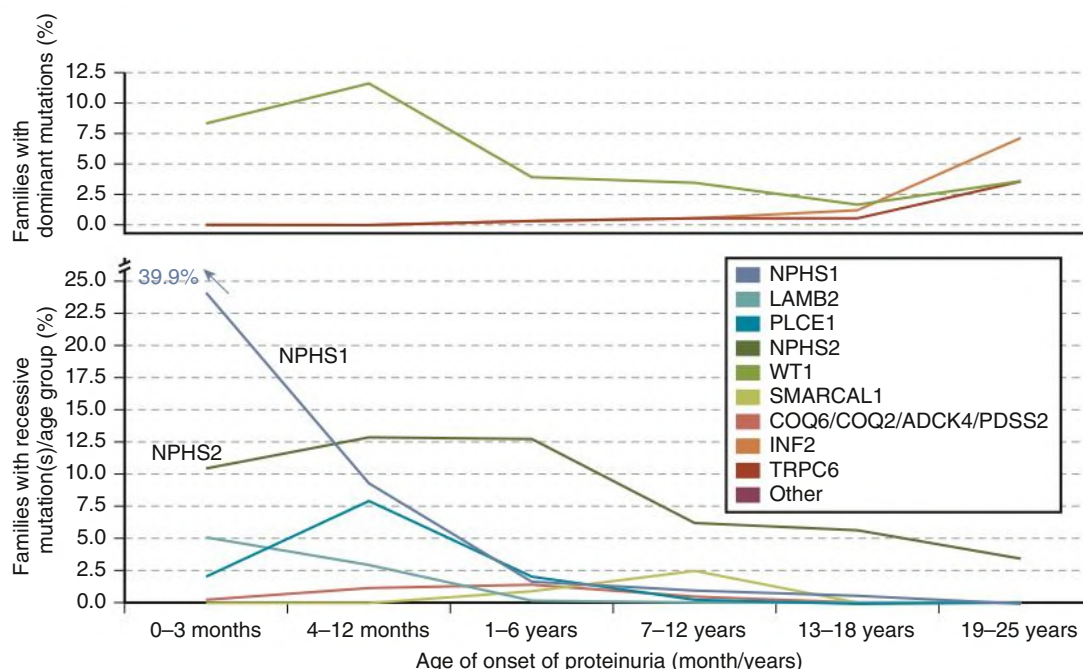


Fig. 20.2 Relative Distribution of Genes that Cause Nephrotic Syndrome, by Age of Onset. Percentages of families in an international cohort with steroid-resistant nephrotic syndrome resulting from mutations in recessive genes are interconnected by lines between age groups and shown in different colors for each causative gene (lower panel for recessive genes, upper panel for dominant genes). *NPHS1* mutations (red), *LAMB2* (orange), and *PLCE1* (dark blue) have early age of onset and are rarely found in patients older than 6 years. The dominant genes *INF2* (light blue) and *TRPC6* (brown) manifest in early adulthood, and *WT1* (black) shows a biphasic distribution with a first peak at 4 to 12 months and a second peak for age of onset beyond 18 years (upper panel). These findings are compatible with the notion that mutations in recessive disease genes are found more frequently in early-onset disease, whereas mutations in dominant genes more frequently cause adult-onset disease. (Modified from Sadowski CE, Lovric S, Ashraf S, et al. A single-gene cause in 29.5% of cases of steroid-resistant nephrotic syndrome. *J Am Soc Nephrol*. 2015;26[6]:1279–1289.)

Detection of a Causal Mutation in 300 Families by Whole-Exome Sequencing

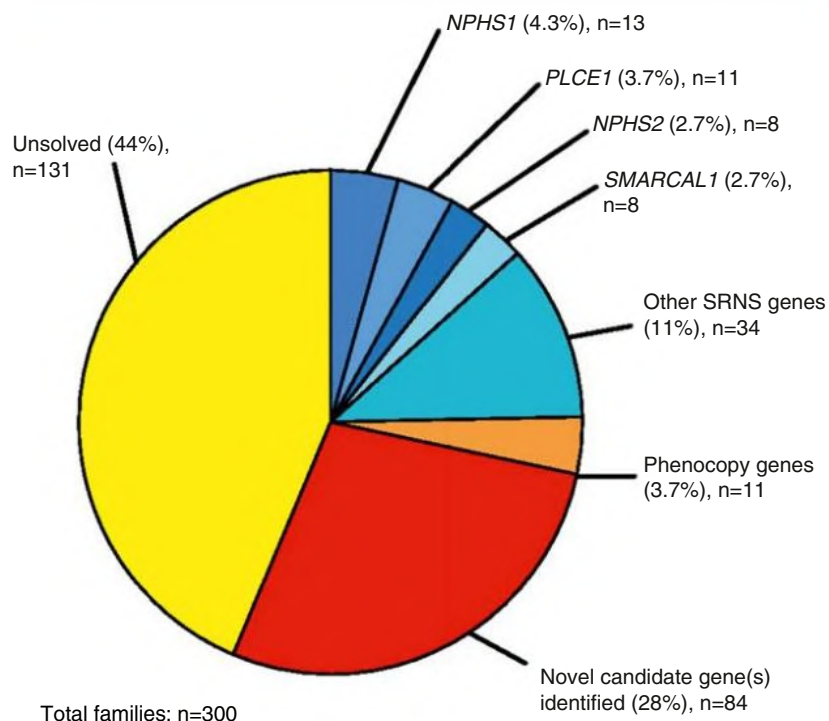


Fig. 20.3 Detection of a Causal Mutation in 300 Families by Whole-Exome Sequencing. In 74 of 300 (25%) families with steroid-resistant nephrotic syndrome (SNRS), a causal mutation was detected in one of 20 genes known to cause steroid-resistant nephrotic syndrome (*shades of blue*). In 3.7% of families, a mutation was found in genes causing a kidney disease that may represent phenocopies of steroid-resistant nephrotic syndrome (*orange*). In 28% of families, one or more potential novel candidate genes were identified (*red*). In 44% of families, no causal mutations or candidate genes were detected. (From Warejko JK, Tan W, Daga A, et al. Whole exome sequencing of patients with steroid-resistant nephrotic syndrome. *Clin J Am Soc Nephrol*. 2018;13[1]:53–62.)

TABLE 20.2 Syndromic Proteinuric Kidney Disease

Syndrome	Gene(s), Mode of Inheritance	CLINICAL FEATURES			
		Kidney Disease	Extrarenal Manifestations	Management	Outcome
AMRF syndrome	<i>SCARB2</i> , AR	Collapsing FSGS (tubular proteinuria was also described) Deposition of C1q and IgM and of C3 in capillary loops and mesangium Adolescent to adulthood age of onset	Progressive myoclonus epilepsy	Supportive	Most affected individuals die in third decade of life
CMTDIE	<i>INF2</i> , AD	SRNS; FSGS; median onset of proteinuria is 18 years	Sensorineural deafness; skeletal abnormalities; peripheral neuropathy with median age of onset of 13 years	Supportive	ESKD develops at median age of 21 years
Denys-Drash	<i>WT1</i> , AD	DMS; high incidence of severe hypertension and rapid progression to ESKD by 3 years of age	Male pseudohermaphroditism, progressive glomerulopathy, development of WT	Bilateral nephrectomy at time of kidney replacement therapy initiation is recommended to prevent WT development/relapse	Dependent on ESKD and WT management Most individuals live to adulthood

Continued

TABLE 20.2 Syndromic Proteinuric Kidney Disease—cont'd

Syndrome	Gene(s), Mode of Inheritance	CLINICAL FEATURES			
		Kidney Disease	Extrarenal Manifestations	Management	Outcome
Frasier	<i>WT1</i> , AD	FSGS; NS may be slowly progressive, usually over 10 years, and is typically steroid resistant	Male pseudohermaphroditism typically present in phenotypic females with amenorrhea or NS, or both; development of genitourinary tumors	Screening for gonadoblastoma; WT not common Evaluate need for an early gonadectomy to prevent tumorigenesis	ESKD development at second to third decade of life Limited information on life expectancy
Galloway-Mowat	<i>WDR73</i> ; <i>WDR4</i> ; <i>KEOPS</i> complex (<i>LAGE3</i> , <i>OSGEP</i> , <i>TP53RK</i> , <i>TPRKB</i> , <i>YRDC</i> and <i>GON7</i>); <i>PRDM15</i> , All: AR	Kidney disease generally presents as NS within the first few months of life, with rapid progression to ESKD	Microcephaly, abnormal cerebral gyral patterns, seizures, psychomotor delay, cranial dysmorphism Most affected children die before age 6 years	Supportive	Most affected individuals die in early childhood
Nail-Patella (hereditary onycho-osteodysplasia; Fong syndrome)	<i>LMX1B</i> , AD	NS, early childhood	Nail dysplasia, absence of or poorly developed patellas, dysplasia of the iliac horns and elbows, cataracts, glaucoma	Treatment of joint limitations, ocular and kidney complications	Clinical expression variable; cases range from mild with no functional impact to severe leading to disability Kidney involvement reported in 30%–60% of cases; life span is normal
Pierson	<i>LAMB2</i> , AR	Congenital NS; DMS on biopsy; rapid progression to ESKD	Hypoplasia of the ciliary body and iris resulting in fixed narrowing of the pupil (microcoria)	Supportive	Most patients progress toward kidney failure within the first year of life Visual prognosis is generally poor
Schimke immuno-osseous dysplasia	<i>SMARCA1</i> , AR	SRNS, FSGS, ESKD	Growth retardation, T cell deficiency, bone dysplasia, cerebrovascular disease	Due to underlying immune disorder, immunosuppressive therapy after kidney transplantation is associated with increased risk of rejection and severe opportunistic infections	Most individuals do not survive beyond early adolescence due to stroke, infections, bone marrow failure, and kidney failure
SPLIS	<i>SGPL1</i> , AR	SRNS; FSGS; typically presents at early childhood (age 1–4 years)	Primary adrenal insufficiency, endocrine abnormalities, immunodeficiency, neurologic abnormalities (cranial nerve deficits, peripheral neuropathy, developmental delay), sensorineural hearing loss, ichthyosis	Multidisciplinary, supportive Recent report ³² suggests potential response to treatment with vitamin B ₆	Overall prognosis is unclear (variable) Risk of death due to infection complications

AD, autosomal dominant; AMRF, action myoclonus-renal failure; AR, autosomal recessive; CMTDIE, Charcot-Marie-Tooth Dominant intermediate E; DMS, diffuse mesangial sclerosis; ESKD, end-stage kidney disease; FSGS, focal segmental glomerulosclerosis; IgM, immunoglobulin M; NS, nephrotic syndrome; SPLIS, Sphingosine phosphate lyase insufficiency syndrome; SRNS, steroid-resistant nephrotic syndrome; WT, Wilms tumor.

coding sequence, is increasingly available, albeit often limited to research initiatives (<https://www.precisionnephrology.org>, <http://www.renalgene.org>). WES allows sequencing of all approximately 330,000 exons in the human genome (i.e., the exome).¹⁶ It is currently assumed that WES offers a theoretical likelihood of 86% of detecting the disease-causing mutation in a recessive disease.¹⁷ Besides its use to detect mutations in an established list of known disease-causing genes, WES was also very successfully applied to detecting novel disease-causing genes (e.g., *ADCK4*, *EMP2*, *CRB2*, *FAT1*, *NUP93*, KEOPS complex, and *TNS2*).^{15,18–21}

The indications for genetic testing in individuals with SRNS are based on several considerations:

1. *Age of onset.* Because of the high likelihood (~30%) of finding a causative monogenic mutation in SRNS with onset before age 25 years,^{4,5} and because of the many important implications for disease management, it is advisable to suggest genetic testing to all individuals with FSGS or with persistent proteinuria that manifests before age 25 years. In adult-onset FSGS, genetic testing should be considered if there is a positive family history^{22,23} or if there is clinical suspicion of a disease syndrome that may mimic SRNS by a phenocopy gene (as can be the case with mutations in type 4 collagen forming genes).^{24,25} Moreover, recent studies suggest a diagnostic yield for WES of up to 17.1% in adults with a clinical diagnosis of nephropathy of unknown origin,²³ and therefore genetic testing should be considered in these cases as well. Finally, individuals of African ancestry with FSGS should be tested for the presence of the *APOL1* risk alleles, at any age of presentation, because mutations in these alleles confer a 10-fold higher risk of end-stage kidney disease (ESKD) due to FSGS and sevenfold higher risk of ESKD due to hypertensive nephropathy.²⁶
2. *Consanguinity.* The likelihood of finding a causative (recessive) mutation is higher in individuals with SRNS from consanguineous marriages than in outbred individuals.^{6,27}
3. *Family history.* One of the most important genotype-phenotype correlations in SRNS is the distinction between recessive versus dominant SRNS genes (see Table 20.1). In recessive mutations, family history is most likely negative because parents of individuals with recessive mutations will usually be healthy heterozygous carriers, and no one in the ancestry will have had the disease. If there is any inherited mutation, it will have only been heterozygous in ancestors. However, in dominant disease, one of the parents of an affected individual will most likely be affected, and the disease may have been handed down through multiple generations (except for situations of incomplete penetrance or de novo mutations). Therefore, the detection of dominant mutations has important clinical implications, such as in situations of planned living-related donor kidney transplantation (see Table 20.1). Here, it will be important to exclude the presence of the disease-causing mutation in the related donor, in whom SRNS may not yet have manifested because of incomplete penetrance, which can be a feature of autosomal dominant inheritance. Kidney donation from a heterozygous carrier of a recessive gene is essentially harmless.
4. *Syndromic features.* The presence of extrarenal manifestations can suggest a mutation in a known gene that results in specific syndrome (see later) and hence increases the likelihood of detecting a monogenic cause of the disease (see Table 20.2).
1. *Response to steroid treatment.* In childhood SRNS, which manifests beyond the first year of life, often one attempt is made at treatment with corticosteroids.²⁸ If a response is lacking, treatment attempts are made with calcineurin inhibitors, and, more recently, mycophenolate mofetil or rituximab. The majority of the patients with NS, in whom a single-gene cause is detected, do not respond to steroid treatment, and therefore the detection of a monogenic cause of SRNS can save unnecessary ineffective and potentially toxic treatment. Recently, however, six genes (*EMP2*, *TNS2*, *DLC1*, *CDK20*, *ITSN1*, *ITSN2*) were identified in which recessive mutations surprisingly cause NS that is partially or fully responsive to steroid therapy or cyclosporine A, for unknown reasons (see Table 20.1).^{3,15,20}
2. *Response to CoQ10 supplementation.* In patients who carry mutations in a gene of CoQ₁₀ biosynthesis (*COQ2*, *COQ6*, *ADCK4*, or *PDSS2*),^{19,29,30} treatment with CoQ₁₀ may be indicated.²⁹
3. *Response to treatment with vitamin B.* The *CUBN* gene encodes the cubilin protein, which functions as a receptor for intrinsic factor–vitamin B₁₂ complexes. Mutations in *CUBN* may benefit the treatment with vitamin B₁₂,³¹ those with sphingosine-1-phosphate lyase 1, encoding a mitochondrial lyase enzyme, may respond to vitamin B₆.³²
4. *Response to other available drugs.* Individuals with *ARHGDI* mutations theoretically may be responsive to eplerenone treatment.³³ Finally, *TRPC6* mutations may potentially be amenable to treatment with calcineurin inhibitors.^{7,34} However, no systematic studies exist for these theoretical treatment modalities.
5. *Other considerations.* Rare syndromic proteinuric kidney diseases require management of extrarenal manifestations (see Table 20.2). For instance, donor splice-site heterozygous mutations in intron 9 of the *WT1* gene have been reported to alter the alternative splicing leading to two *WT1* isoforms, with (+) or without (–) three amino acids, lysine-threonine-serine (KTS), between zinc fingers three and four. The detection of *WT1* mutations often has clinical consequences as, for instance, KTS+ mutations, depending on karyotype, may confer a risk for gonadoblastoma.³⁵

RECURRENCE

Childhood-onset SRNS carries a risk of about 33% for relapse in a kidney transplant, thereby causing recurrence of CKD³⁶ (see Chapter 113). However, kidney transplantation in a child with a known monogenic cause of SRNS carries almost no risk of disease recurrence because the defective gene, which is usually exclusively expressed in the kidney, is now replaced with a normal one. A case of a truncating mutation (resulting in no formation of a functional protein) and the introduction of a novel protein through the donor kidney may rarely elicit an antibody response (such as in Alport posttransplant nephritis) and rejection of the transplanted organ.³⁷ However, posttransplant immunosuppressive regimens probably protect against such a response in most cases (see Chapter 113).

FUTURE DIRECTIONS

Currently only “strong” mutations are considered as disease causing (see Box 20.1). However, it is very likely that a high percentage of adult-onset SRNS is caused by hypomorphic (“weak”) recessive alleles, which have not yet been revealed as deleterious (e.g., the allele R229Q of *NPHS2*). One of the most important tasks in the future for kidney geneticists is to use cell-based and animal model systems to explore whether “weak” recessive mutations (e.g., one hypomorphic heterozygous variant in a compound-heterozygous mutation) may cause adult-onset SRNS.

The recent identification of monogenic causes of NS that manifests before age 25 years in as many as 12% to 45% of individuals

Natural History and Management

The identification of causative monogenic mutations may have important therapeutic consequences in some cases:

TABLE 20.3 Implications of the Identification of Causative Mutations in Nephrotic Syndrome

Implication	Significance ^a
Unequivocal molecular genetic diagnosis	Avoidance of unnecessary diagnostic procedures; positive psychological impact on the individual and family members
Prediction of treatment response	Avoidance of steroids/immunosuppressive therapy
Establishment of genotype–phenotype correlations	Can assist in management (anticipated extrarenal manifestations or complications); etiologic stratification of participants for therapeutic studies by specific causative gene and mutation
Study of mutations' detrimental effects using genetic animal models	Study the pathogenesis of steroid-resistant nephrotic syndrome as the basis for targeted therapy development
Discovery of distinct mutations that may be amenable to treatment	May offer a curative therapy
Inform clinical decisions	Guide the identity of a living related donor in dominant diseases

^aSee also <https://www.precisionnephrology.org>.

worldwide⁴ has offered many advantages for future management of NS. With the available sequencing technology and the continuous reduction in sequencing cost, WES on a commercial or research basis should now be offered to every patient with persistent proteinuria occurring before age 25 years, or in patients with FSGS manifesting after age 25 years and positive family history, if the patient consents for clinical genetic testing for the following reasons (Table 20.3): (1) it will provide the patient and families with an unequivocal diagnosis; (2) it will establish genotype–phenotype correlations; (3) it may avoid the need for kidney biopsy; (4) it will further unravel the puzzle of pathogenic pathways of NS and enable etiologic stratification of participants for therapeutic studies by specific causative gene and mutation; (5) it will permit personalized treatment options for NS by discovery of distinct mutations that may be amenable to treatment (e.g., CoQ₁₀-related genes), or on the other hand, support a decision to avoid immunosuppression or corticosteroids; and (6) it will help establish informed clinical decisions (e.g., in kidney transplantation) (see <https://www.precisionnephrology.org>).³⁸

SELF-ASSESSMENT QUESTIONS

1. A 23-year-old man is admitted to the hemodialysis unit after inpatient hemodialysis initiation for ESKD secondary to FSGS. At initial evaluation, you obtain a detailed family history in which the patient discloses that his maternal grandfather, mother, and two brothers have received dialysis. You suspect familial FSGS as the cause of renal disease in this family. Based on the pattern of kidney disease occurrence in his family, what mode of inheritance would you suspect in this patient?
 - A. Autosomal recessive
 - B. X-linked recessive
 - C. Autosomal dominant
 - D. X-linked dominant
 - E. Both C and D are possible
2. You are called to consult on a patient with newly diagnosed FSGS. Her family history is significant for multiple family members with advanced CKD or ESKD requiring renal replacement therapy. Three family members are known to have had renal transplants without recurrence of disease. You construct a family pedigree that suggests an autosomal dominant inheritance pattern. Of the known genetic mutations to cause autosomal dominant FSGS, which gene may be affected in this patient?
 - A. *NPHS1*
 - B. *MYO1E*
 - C. *INF2*
 - D. *NPHS2* (podocin)
3. Genetic mutations affecting which of the following cell populations have been identified as a cause of familial FSGS?
 - A. Glomerular epithelial
 - B. Glomerular endothelial
 - C. Glomerular mesangial
 - D. Tubular epithelial
4. What is the approximate likelihood of finding a monogenic cause of steroid-resistant nephrotic syndrome that manifests by age 25 years?
 - A. 5%
 - B. 30%
 - C. 50%
 - D. 80%
5. You are consulted on a 5-year-old girl with significant nephrotic range proteinuria. On genetic analysis, she is found to have a homozygous missense mutation in the *COQ6* gene. Which of the following would be the most appropriate treatment that could help improve her condition?
 - A. Steroids
 - B. Coenzyme Q₁₀
 - C. Eplerenone
 - D. Calcineurin inhibitors

REFERENCES

- Kestila M, Lenkkeri U, Mannikko M, et al. Positionally cloned gene for a novel glomerular protein-nephrin is mutated in congenital nephrotic syndrome. *Mol Cell*. 1998;1(4):575–582.
- Boute N, Gribouval O, Roselli S, et al. NPHS2, encoding the glomerular protein podocin, is mutated in autosomal recessive steroid-resistant nephrotic syndrome. *Nat Genet*. 2000;24(4):349–354.
- Hinkes B, Wiggins RC, Gbadegesin R, et al. Positional cloning uncovers mutations in PLCE1 responsible for a nephrotic syndrome variant that may be reversible. *Nat Genet*. 2006;38(12):1397–1405.
- Sadowski CE, Lovric S, Ashraf S, et al. A single-gene cause in 29.5% of cases of steroid-resistant nephrotic syndrome. *J Am Soc Nephrol*. 2015;26(6):1279–1289.
- Hildebrandt F, Heeringa SF. Specific podocin mutations determine age of onset of nephrotic syndrome all the way into adult life. *Kidney Int*. 2009;75(7):669–671.
- Warejko JK, Tan W, Daga A, et al. Whole exome sequencing of patients with steroid-resistant nephrotic syndrome. *Clin J Am Soc Nephrol*. 2018;13(1):53–62.
- Winn MP, Conlon PJ, Lynn KL, et al. A mutation in the TRPC6 cation channel causes familial focal segmental glomerulosclerosis. *Science*. 2005;308(5729):1801–1804.
- Brown EJ, Schlondorff JS, Becker DJ, et al. Mutations in the formin gene INF2 cause focal segmental glomerulosclerosis. *Nat Genet*. 2010;42(1):72–76.
- Benoit G, Machuca E, Heidet L, Antignac C. Hereditary kidney diseases: highlighting the importance of classical Mendelian phenotypes. *Ann N Y Acad Sci*. 2010;1214:83–98.
- Diomedè-Camassei F, Di Giandomenico S, Santorelli FM, et al. COQ2 nephropathy: a newly described inherited mitochondriopathy with primary renal involvement. *J Am Soc Nephrol*. 2007;18(10):2773–2780.
- Koziell A, Grech V, Hussain S, et al. Genotype/phenotype correlations of NPHS1 and NPHS2 mutations in nephrotic syndrome advocate a functional inter-relationship in glomerular filtration. *Hum Mol Genet*. 2002;11(4):379–388.
- Hinkes B, Vlangos C, Heeringa S, et al. Specific podocin mutations correlate with age of onset in steroid-resistant nephrotic syndrome. *J Am Soc Nephrol*. 2008;19(2):365–371.
- Tory K, Menyhard DK, Woerner S, et al. Mutation-dependent recessive inheritance of NPHS2-associated steroid-resistant nephrotic syndrome. *Nat Genet*. 2014;46(3):299–304.
- Boyer O, Nevo F, Plaisier E, et al. INF2 mutations in Charcot-Marie-Tooth disease with glomerulopathy. *N Engl J Med*. 2011;365(25):2377–2388.
- Ashraf S, Kudo H, Rao J, et al. Mutations in six nephrosis genes delineate a pathogenic pathway amenable to treatment. *Nat Commun*. 2018;9(1):1960.
- Teer JK, Mullikin JC. Exome sequencing: the sweet spot before whole genomes. *Hum Mol Genet*. 2010;19(R2):R145–R151.
- Lifton RP. Individual genomes on the horizon. *N Engl J Med*. 2010;362(13):1235–1236.
- Braun DA, Rao J, Mollet G, et al. Mutations in KEOPS-complex genes cause nephrotic syndrome with primary microcephaly. *Nat Genet*. 2017;49(10):1529–1538.
- Ashraf S, Gee HY, Woerner S, et al. ADCK4 mutations promote steroid-resistant nephrotic syndrome through CoQ10 biosynthesis disruption. *J Clin Invest*. 2013;123(12):5179–5189.
- Gee HY, Ashraf S, Wan X, et al. Mutations in EMP2 cause childhood-onset nephrotic syndrome. *Am J Hum Genet*. 2014;94(6):884–890.
- Braun DA, Sadowski CE, Kohl S, et al. Mutations in nuclear pore genes NUP93, NUP205 and XPO5 cause steroid-resistant nephrotic syndrome. *Nat Genet*. 2016;48(4):457–465.
- Connaughton DM, Kennedy C, Shril S, et al. Monogenic causes of chronic kidney disease in adults. *Kidney Int*. 2019;95(4):914–928.
- Groopman E, Goldstein D, Gharavi A. Diagnostic utility of exome sequencing for kidney disease. Reply. *N Engl J Med*. 2019;380(21):2080–2081.
- Braunisch MC, Buttner-Herold M, Gunthner R, et al. Heterozygous COL4A3 variants in histologically diagnosed focal segmental glomerulosclerosis. *Front Pediatr*. 2018;6:171.
- Schapiro D, Daga A, Lawson JA, et al. Panel sequencing distinguishes monogenic forms of nephritis from nephrosis in children. *Nephrol Dial Transplant*. 2019;34(3):474–485.
- Genovese G, Friedman DJ, Ross MD, et al. Association of trypanolytic ApoL1 variants with kidney disease in African Americans. *Science*. 2010;329(5993):841–845.
- Hildebrandt F, Heeringa SF, Ruschendorf F, et al. A systematic approach to mapping recessive disease genes in individuals from outbred populations. *PLoS Genet*. 2009;5(1):e1000353.
- Nephrotic syndrome in children: prediction of histopathology from clinical and laboratory characteristics at time of diagnosis. a report of the International Study of Kidney Disease in Children. *Kidney Int*. 1978;13(2):159–165.
- Heeringa SF, Chernin G, Chaki M, et al. COQ6 mutations in human patients produce nephrotic syndrome with sensorineural deafness. *J Clin Invest*. 2011;121(5):2013–2024.
- Atmaca M, Gulhan B, Atayar E, et al. Long-term follow-up results of patients with ADCK4 mutations who have been diagnosed in the asymptomatic period: effects of early initiation of CoQ10 supplementation. *Turk J Pediatr*. 2019;61(5):657–663.
- Ovunc B, Otto EA, Vega-Warner V, et al. Exome sequencing reveals cubilin mutation as a single-gene cause of proteinuria. *J Am Soc Nephrol*. 2011;22(10):1815–1820.
- Zhao P, Liu ID, Hodgins JB, et al. Responsiveness of sphingosine phosphate lyase insufficiency syndrome to vitamin B₆ cofactor supplementation. *J Inher Metab Dis*. 2020;43(5):1131–1142.
- Gee HY, Saisawat P, Ashraf S, et al. ARHGDI4 mutations cause nephrotic syndrome via defective RHO GTPase signaling. *J Clin Invest*. 2013;123(8):3243–3253.
- Reiser J, Polu KR, Moller CC, et al. TRPC6 is a glomerular slit diaphragm-associated channel required for normal renal function. *Nat Genet*. 2005;37(7):739–744.
- Chernin G, Vega-Warner V, Schoeb DS, et al. Genotype/phenotype correlation in nephrotic syndrome caused by WT1 mutations. *Clin J Am Soc Nephrol*. 2010;5(9):1655–1662.
- Hildebrandt F. Genetic kidney diseases. *Lancet*. 2010;375(9722):1287–1295.
- Dehan P, Van den Heuvel LP, Smeets HJ, Tryggvason K, Foidart JM. Identification of post-transplant anti-alpha 5 (IV) collagen alloantibodies in X-linked Alport syndrome. *Nephrol Dial Transplant*. 1996;11(10):1983–1988.
- Mann N, Braun DA, Amann K, et al. Whole-exome sequencing enables a precision medicine approach for kidney transplant recipients. *J Am Soc Nephrol*. 2019;30(2):201–215.
- Jeanpierre C, Denamur E, Cabanis HMO, et al. Identification of constitutional WT1 mutations, in patients with isolated diffuse mesangial sclerosis, and analysis of genotype/phenotype correlations by use of a computerized mutation database. *Am J Hum Genet*. 1998;62:824–833.
- Kaplan JM, Kim SH, North KN, et al. Mutations in ACTN4, encoding alpha-actinin-4, cause familial focal segmental glomerulosclerosis. *Nat Genet*. 2000;24:215–216.
- Heath KE, Campos-Barros A, Toren A, et al. Nonmuscle myosin heavy chain IIA mutations define a spectrum of autosomal dominant macrothrombocytopenias: may-Hegglin anomaly and Fechtner, Sebastian, Epstein, and Alport-like syndromes. *Am J Hum Genet*. 2001;69:1033–1045.
- Barua M, Stellacci E, Stella L, et al. Mutations in PAX2 associate with adult-onset FSGS. *J Am Soc Nephrol*. 2014;25:1942–1953.
- Gbadegesin RA, Hall G, Adeyemo A, et al. Mutations in the gene that encodes the F-actin binding protein anillin cause FSGS. *J Am Soc Nephrol*. 2014;25:1991–2002.
- Zenker M, Aigner T, Wendler O, et al. Human laminin beta2 deficiency causes congenital nephrosis with mesangial sclerosis and distinct eye abnormalities. *Hum Mol Genet*. 2004;13:2625–2632.
- Boerkoel CF, Takashima H, John J, et al. Mutant chromatin remodeling protein SMARCA1 causes Schimke immuno-osseous dysplasia. *Nat Genet*. 2002;30:215–220.

46. Dreyer SD, Zhou G, Baldini A, et al. Mutations in LMX1B cause abnormal skeletal patterning and renal dysplasia in nail patella syndrome. *Nat Genet.* 1998;19:47–50.
47. Heeringa SF, Chernin G, Moumita C, et al. COQ6 mutations in human patients produce nephrotic syndrome with sensorineural deafness. *J Clin Invest.* 2011;121:2013–2024.
48. Ozaltin F, Li B, Rauhauser A, et al. DGKE variants cause a glomerular microangiopathy that mimics membranoproliferative GN. *J Am Soc Nephrol.* 2013;24:377–384.
49. Gee HY, Sadowski CE, Aggarwal PK, et al. FAT1 mutations cause a glomerulotubular nephropathy. *Nat Commun.* 2016;7:10822.
50. Lahrouchi N, George A, Ratbi I, et al. Homozygous frameshift mutations in FAT1 cause a syndrome characterized by colobomatous-microphthalmia, ptosis, nephropathy and syndactyly. *Nat Commun.* 2019;10:1180.
51. Bierzynska A, Soderquest K, Dean P, et al. MAGI2 mutations cause congenital nephrotic syndrome. *J Am Soc Nephrol.* 2017;28:1614–1621.
52. Weng PL, Majmundar AJ, Khan K, et al. De novo TRIM8 variants impair its protein localization to nuclear bodies and cause developmental delay, epilepsy, and focal segment glomerulosclerosis. *Am J Hum Genet.* 2021;108:357–367.